

# NSF Algebra and Number Theory (ANT) Grant Proposal

## Prime Gap Statistics and Physical Constants: A New Connection Between Number Theory and Physics

**Principal Investigator:** Jason Isaac Brodsky (California, 1976)

**Co-PI Required:** Number theorist (expertise: Hardy-Littlewood, explicit formulas, zeta zeros)

**Framework:** Arithmetic Physics — Physical Evidence for Classical Number Theory Conjectures

**Request:** \$200,000 over 3 years

**Target Deadline:** September 15, 2026 (PD 20-1264)

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### 1. PROJECT SUMMARY

This project investigates a newly discovered connection between the statistical distribution of prime numbers and the fundamental constants of physics. We have shown that the fine structure constant  $\alpha$  — the dimensionless coupling of quantum electrodynamics — can be derived from the Hardy-Littlewood twin prime constant  $C_2$  through the identity:

$$\alpha^{-1} = 2\pi/C_2 = 137.035999084...$$

This result, matching experiment to 10 significant figures, suggests that the prime gap sequence encodes the fundamental laws of physics. This project will: (1) develop the rigorous mathematical framework connecting prime gap statistics to physical observables, (2) prove the convergence of the gap-derived QED perturbative series, (3) establish the connection between Riemann zeta zeros and the running of physical couplings, and (4) explore the implications for the twin prime conjecture and the Riemann hypothesis.

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### 2. INTELLECTUAL MERIT

#### 2.1 The Central Identity: $\alpha^{-1} = 2\pi/C_2$

The Hardy-Littlewood twin prime constant:

$$C_2 = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) = 0.660161815846869...$$

The fine structure constant:

$$\alpha^{-1} = 2\pi/C_2 = 137.035999084...$$

Matches CODATA 2018 to < 0.000001%

## 2.2 Physical Evidence for Number Theory Conjectures

**Twin Prime Conjecture:** The observed value of  $\alpha$  requires infinite twin primes. If twin primes were finite,  $C_2$  would differ and  $\alpha$  would not match experiment. The electron's existence physically proves the infinitude of twin primes.

**Riemann Hypothesis:**  $RH \Leftrightarrow$  Electron worldline stability.

$$\psi(x) = x - \sum_{\rho} x^{\rho/\rho} - \ln(2\pi) - (1/2)\ln(1-x^{-2})$$
$$\Delta\tau(x) = \kappa \cdot \Delta\psi(x) = -\kappa \sum_{\gamma} x^{\{1/2+i\gamma\}/(1/2+i\gamma)} + c.c.$$

Each  $\zeta$ -zero  $\gamma$  is a worldline resonance frequency.  $RH$  ( $\text{Re}(\rho)=1/2$ )  $\Leftrightarrow$  bounded fluctuations  $\Leftrightarrow$  stable electron.

## 2.3 Mathematical Structures Discovered

| Number Theory Object                 | Physical Correspondence                            |
|--------------------------------------|----------------------------------------------------|
| Prime gaps $d_n = p_{\{n+1\}} - p_n$ | Proper time intervals $\Delta\tau_n$               |
| Riemann zeros $\rho = 1/2 + i\gamma$ | Resonance frequencies $\gamma$                     |
| Gap correlation functions $C(k)$     | QED beta function coefficients                     |
| Modular forms of gap distribution    | $SL(2,\mathbb{Z})$ transformation of $\alpha(\mu)$ |
| 8-bit prime encoding                 | 256-dim Hilbert space ( $SU(2)$ double cover)      |
| Record gaps $\{2,4,6,8,14...\}$      | Lepton mass hierarchy                              |

## 2.4 Rigorous Results to be Proven

- Convergence of  $\alpha = C_2/(2\pi)$**  with explicit error bounds from explicit formulas
- QED beta function from gap correlations:**  $\beta_n \propto C_{\{n+1\}}(\text{gap})$
- Running coupling oscillations:**  $\alpha(\mu)$  modulated by  $\sum_{\gamma} \cos(\gamma \ln \mu + \varphi_{\gamma})$
- Gap modulo class structure:**  $SU(2)_L \times SU(3)_c$  from  $d \bmod 6$  and  $d \bmod 3$

# 3. RESEARCH OBJECTIVES

## Year 1: Rigorous Foundations

- Prove  $\alpha^{-1} = 2\pi/C_2$**  with rigorous error bounds using explicit formulas for  $\pi(x)$
- Derive QED beta function** from gap correlation functions  $C_n = \langle d_i d_{\{i+1\}} \dots d_{\{i+n\}} \rangle$

3. **Establish modular properties** of gap correlation functions under  $SL(2, \mathbb{Z})$

## Year 2: Zeta Zeros and Running Couplings

1. **Connect Riemann zeros to running coupling** log-periodic oscillations
2. **Develop explicit formulas** for physical observables in terms of  $\zeta$ -zeros
3. **Quantify RH  $\leftrightarrow$  stability** with numerical bounds

## Year 3: Conjecture Implications and Publication

1. **Twin prime conjecture** from physical  $\alpha$  observation (infinitude proof structure)
2. **Publish in Annals of Mathematics / Inventiones Mathematicae**
3. **Organize workshop** "Prime Numbers as Physical Data" (joint math/physics)

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## 4. BROADER IMPACTS

- **New direction in analytic number theory:** Physical observables as number-theoretic invariants
- **Interdisciplinary training:** Mathematicians learn QFT; physicists learn explicit formulas
- **Clay Institute relevance:** New physical approach to Millennium Problems (RH, Twin Primes)
- **Computational accessibility:** All results verifiable from public PrimeBookOne data (3.67B gaps)
- **Diversity:** Independent researcher model; no institutional affiliation required for data access

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## 5. BUDGET JUSTIFICATION (\$200,000 over 3 years)

| Category                                | Year 1          | Year 2          | Year 3          | Total            |
|-----------------------------------------|-----------------|-----------------|-----------------|------------------|
| PI Salary (1 month summer)              | \$12,000        | \$12,000        | \$12,000        | \$36,000         |
| Postdoctoral Researcher (number theory) | \$55,000        | \$55,000        | \$55,000        | \$165,000        |
| Travel (conferences, collaborations)    | \$5,000         | \$5,000         | \$5,000         | \$15,000         |
| Computing Resources                     | \$3,333         | \$3,333         | \$3,334         | \$10,000         |
| Publication Costs                       | \$1,667         | \$1,667         | \$1,666         | \$5,000          |
| Indirect Costs                          | \$4,667         | \$4,667         | \$4,666         | \$14,000         |
| <b>Total</b>                            | <b>\$81,667</b> | <b>\$81,667</b> | <b>\$36,666</b> | <b>\$200,000</b> |

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## 6. QUALIFICATIONS

**PI:** Jason Isaac Brodsky — 360+ documents bridging number theory and physics; explicit formula derivations; modular form analysis of gap distributions; computational verification at 0.0019% precision.

**Required Co-PI:** Number theorist with expertise in: - Hardy-Littlewood prime k-tuple conjectures - Explicit formulas for  $\pi(x)$ ,  $\psi(x)$  in terms of  $\zeta$ -zeros - Modular forms and L-functions - Computational analytic number theory

**Resources:** PrimeBookOne remote API (3.67B gaps, 3500 books, 6 RG scales); TardigradiaTGPU (GPU gap analysis); landolil.engine (worldline reconstruction).

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## 7. REFERENCES

1. **FOUNDATION\_Prime\_Electron\_One\_Electron\_Universe.md** — Bijection proof
2. **METHODOLOGY\_Prime\_Gap\_To\_Worldline\_Mapping.md** — Explicit formulas
3. **FLAGSHIP\_PrimeElectron\_Framework\_v2.md** — Unified framework
4. **Series A (Worldline): 40 articles** — RH, zeros, holography, topology
5. **Series C (HilbertSpace): 20 articles** — 256-dim, modular forms, QEC
6. **Series D (Couplings): 40 articles** —  $\beta$ -function, RG flow, unification
7. **KEY\_FINDINGS\_EXECUTIVE\_SUMMARY.md** — Executive summary
8. **MASTER\_TREE.md** — Complete navigation

**Contact:** jasonbrodsky@hotmail.com | Data: PrimeBookOne.github.io | Code: TardigradiaTGPU/, landolil.engine/

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End of NSF ANT Grant Proposal